Haciendo la guía detallada.

```
vagrant@ubuntu-jammy: ~

PS C:\Users\pablo\UTN-DevOps\utn-devops> vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Checking if box 'ubuntu/jammy64' version '20241002.0.0' is up to date...
==> default: Resuming suspended VM...
==> default: Booting VM...
==> default: Waiting for machine to boot. This may take a few minutes...
    default: SSH address: 127.0.0.1:2222
    default: SSH username: vagrant
    default: SSH auth method: private key
    default:
    default: Vagrant insecure key detected. Vagrant will automatically replace
    default: this with a newly generated keypair for better security.
    default:
    default: Inserting generated public key within guest...
    default: Removing insecure key from the guest if it's present...
    default: Key inserted! Disconnecting and reconnecting using new SSH key...
==> default: Machine booted and ready!
==> default: Running provisioner: shell...
    default: Running: inline script
    default: Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
    default: Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
    default: Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
    default: Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3058 kB]
    default: Get:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
    default: Get:6 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
    default: Get:7 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [434 kB]
    default: Get:8 http://security.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadata [14.1 kB]
    default: Get:9 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [5343 kB]
    default: Get:10 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [1022 kB]
    default: Get:11 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 c-n-f Metadata [680 B
    default: Get:12 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1019 kB]
    default: Get:13 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [226 kB]
```

```
vagrant@ubuntu-jammy: ~

PS C:\Users\pablo\UTN-DevOps\utn-devops> vagrant ssh
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-171-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Mar 18 14:51:00 UTC 2026

  System load:             0.0
  Usage of /:              10.2% of 38.70GB
  Memory usage:            23%
  Swap usage:              0%
  Processes:               106
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:b3:94ff:feb0:5438


Expanded Security Maintenance for Applications is not enabled.

37 updates can be applied immediately.
35 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


vagrant@ubuntu-jammy:~$
```

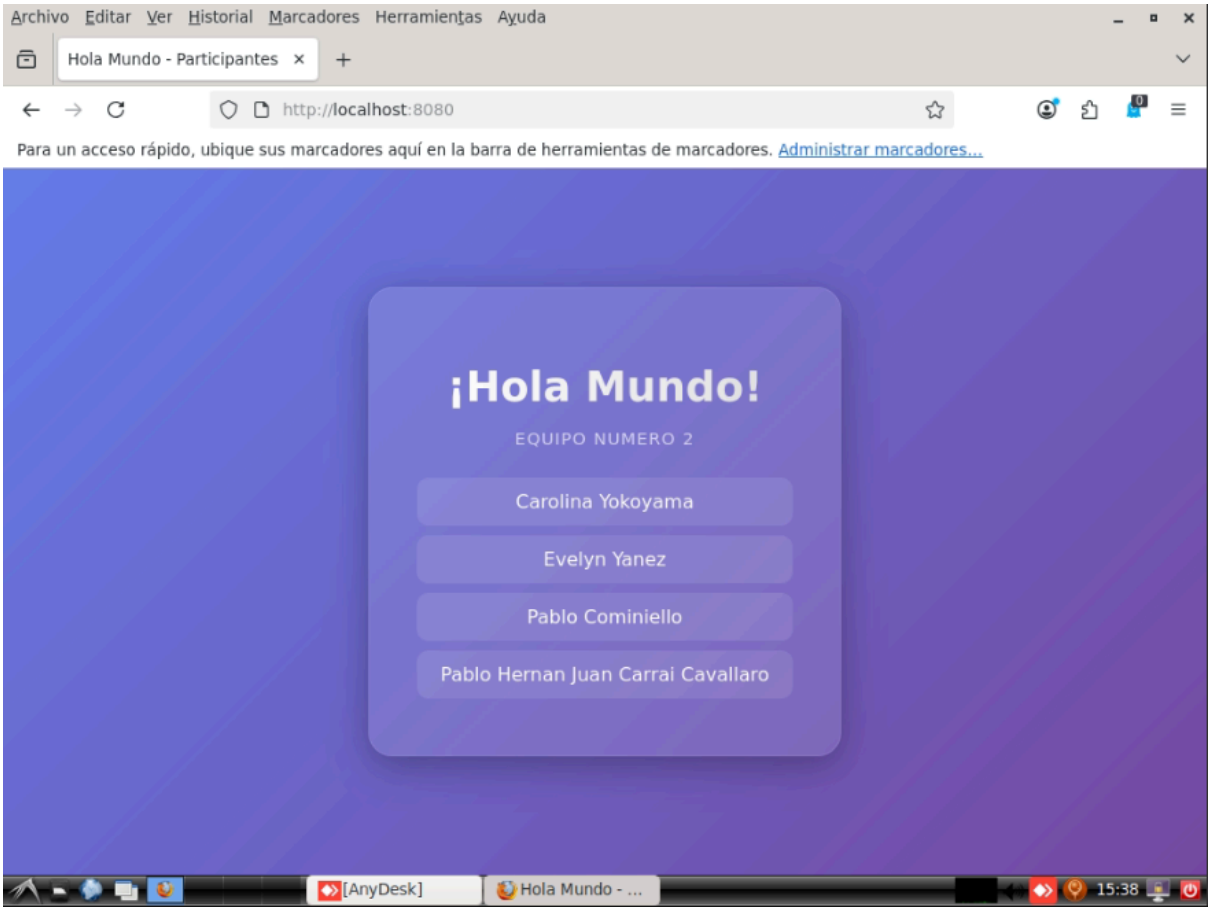
Haciendo la guía exploratoria.

```
ed@fua:~$  git clone https://github.com/PabloCarrai/vagrant-ejemplos.git
Clonando en 'vagrant-ejemplos'...
remote: Enumerating objects: 30, done.
remote: Counting objects: 100% (30/30), done.
remote: Compressing objects: 100% (23/23), done.
remote: Total 30 (delta 7), reused 27 (delta 6), pack-reused 0 (from 0)
Recibiendo objetos: 100% (30/30), 5.32 KiB | 362.00 KiB/s, listo.
Resolviendo deltas: 100% (7/7), listo.
ed@fua:~$ cd vagrant-ejemplos/
ed@fua:~/vagrant-ejemplos$ vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Importing base box 'alvistack/debian-13'...
==> default: Matching MAC address for NAT networking...
==> default: Checking if box 'alvistack/debian-13' version '20251010.1.3' is up to date...
==> default: Setting the name of the VM: vagrant-ejemplos_default_1773945352381_16807
==> default: Clearing any previously set network interfaces...
==> default: Preparing network interfaces based on configuration...
    default: Adapter 1: nat
==> default: Forwarding ports...
    default: 8080 (guest) => 8080 (host) (adapter 1)
    default: 22 (guest) => 2222 (host) (adapter 1)
==> default: Running 'pre-boot' VM customizations...
==> default: Booting VM...
==> default: Waiting for machine to boot. This may take a few minutes...
    default: SSH address: 127.0.0.1:2222
    default: SSH username: vagrant
    default: SSH auth method: private key
    default:
    default: Vagrant insecure key detected. Vagrant will automatically replace
    default: this with a newly generated keypair for better security.
    default:
    default: Inserting generated public key within guest...
    default: Removing insecure key from the guest if it's present...
    default: Key inserted! Disconnecting and reconnecting using new SSH key...
==> default: Machine booted and ready!
==> default: Checking for guest additions in VM...
==> default: Mounting shared folders...
    default: /home/ed/vagrant-ejemplos => /vagrant
==> default: Running provisioner: shell...
    default: Running: inline script
    default: Get:1 https://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
    default: Get:2 https://deb.debian.org/debian trixie InRelease [140 kB]
    default: Get:3 https://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
    default: Get:4 https://deb.debian.org/debian-security trixie-security/main amd64 Packages [115 kB]
    default: Get:5 https://deb.debian.org/debian-security trixie-security/main Translation-en [74.1 kB]
    default: Get:6 https://deb.debian.org/debian-security trixie-security/non-free-firmware amd64 Packages [544 B]
    default: Get:7 https://deb.debian.org/debian-security trixie-security/non-free-firmware Translation-en [352 B]
    default: Get:8 https://deb.debian.org/debian trixie/main amd64 Packages [9,671 kB]
    default: Get:9 https://deb.debian.org/debian trixie/main Translation-en [6,484 kB]
    default: Get:10 https://deb.debian.org/debian trixie/main amd64 Components [5,175 kB]
    default: Get:11 https://deb.debian.org/debian trixie/main Icons (48x48) [3,626 kB]
    default: Get:12 https://downloadcontent.opensuse.org/repositories/home:/alvistack/Debian_13  InRelease [1,576 B]
    default: Err:12 https://downloadcontent.opensuse.org/repositories/home:/alvistack/Debian_13  InRelease
    default:   Sub-process /usr/bin/sqv returned an error code (1), error message is: Signing key on 789CFFDE0295B8A1F4E5690C4BECC97550D0B1FD is bad:
           The primary key is not live   because: Expired on 2025-11-22T06:39:39Z
    default: Get:13 https://deb.debian.org/debian trixie/main Icons (64x64) [7,367 kB]
    default: Get:14 https://deb.debian.org/debian trixie/contrib amd64 Packages [53.8 kB]
    default: Get:15 https://deb.debian.org/debian trixie/contrib Translation-en [49.6 kB]
    default: Get:16 https://deb.debian.org/debian trixie/contrib amd64 Components [41.5 kB]
    default: Get:17 https://deb.debian.org/debian trixie/contrib Icons (48x48) [60.1 kB]
    default: Get:18 https://deb.debian.org/debian trixie/contrib Icons (64x64) [132 kB]
    default: Get:19 https://deb.debian.org/debian trixie/non-free amd64 Packages [100 kB]
    default: Get:20 https://deb.debian.org/debian trixie/non-free Translation-en [67.1 kB]
    default: Get:21 https://deb.debian.org/debian trixie/non-free amd64 Components [3,784 B]
    default: Get:22 https://deb.debian.org/debian trixie/non-free Icons (48x48) [578 B]
    default: Get:23 https://deb.debian.org/debian trixie/non-free Icons (64x64) [10.4 kB]
    default: Get:24 https://deb.debian.org/debian trixie/non-free-firmware amd64 Packages [6,884 B]
    default: Get:25 https://deb.debian.org/debian trixie/non-free-firmware Translation-en [4,704 B]
    default: Get:26 https://deb.debian.org/debian trixie/non-free-firmware amd64 Components [24.3 kB]
    default: Get:27 https://deb.debian.org/debian trixie/non-free-firmware Icons (48x48) [29 B]
    default: Get:28 https://deb.debian.org/debian trixie/non-free-firmware Icons (64x64) [29 B]
    default: Get:29 https://deb.debian.org/debian trixie-updates/main amd64 Packages [5,412 B]
    default: Get:30 https://deb.debian.org/debian trixie-updates/main Translation-en [4,096 B]
    default: Reading package lists...
```

```
    default: dpkg-preconfigure: unable to re-open stdin: No such file or directory
    default: Fetched 13.1 MB in 1s (9,389 kB/s)
(Reading database ... 189246 files and directories currently installed.)
    default: Preparing to unpack .../0-apache2-bin_2.4.66-1~deb13u2_amd64.deb ...
    default: Unpacking apache2-bin (2.4.66-1~deb13u2) over (2.4.65-2) ...
    default: Selecting previously unselected package apache2-data.
    default: Preparing to unpack .../1-apache2-data_2.4.66-1~deb13u2_all.deb ...
    default: Unpacking apache2-data (2.4.66-1~deb13u2) ...
    default: Selecting previously unselected package apache2-utils.
    default: Preparing to unpack .../2-apache2-utils_2.4.66-1~deb13u2_amd64.deb ...
    default: Unpacking apache2-utils (2.4.66-1~deb13u2) ...
    default: Selecting previously unselected package apache2.
    default: Preparing to unpack .../3-apache2_2.4.66-1~deb13u2_amd64.deb ...
    default: Unpacking apache2 (2.4.66-1~deb13u2) ...
    default: Selecting previously unselected package liberror-perl.
    default: Preparing to unpack .../4-liberror-perl_0.17030-1_all.deb ...
    default: Unpacking liberror-perl (0.17030-1) ...
    default: Selecting previously unselected package git-man.
    default: Preparing to unpack .../5-git-man_1%3a2.47.3-0+deb13u1_all.deb ...
    default: Unpacking git-man (1:2.47.3-0+deb13u1) ...
    default: Selecting previously unselected package git.
    default: Preparing to unpack .../6-git_1%3a2.47.3-0+deb13u1_amd64.deb ...
    default: Unpacking git (1:2.47.3-0+deb13u1) ...
    default: Setting up apache2-bin (2.4.66-1~deb13u2) ...
    default: Setting up liberror-perl (0.17030-1) ...
    default: Setting up git-man (1:2.47.3-0+deb13u1) ...
    default: Setting up apache2-data (2.4.66-1~deb13u2) ...
    default: Setting up apache2-utils (2.4.66-1~deb13u2) ...
    default: Setting up apache2 (2.4.66-1~deb13u2) ...
    default: Enabling module mpm_event.
    default: Enabling module authz_core.
    default: Enabling module authz_host.
    default: Enabling module authn_core.
    default: Enabling module auth_basic.
    default: Enabling module access_compat.
    default: Enabling module authn_file.
    default: Enabling module authz_user.
    default: Enabling module alias.
    default: Enabling module dir.
    default: Enabling module autoindex.
    default: Enabling module env.
    default: Enabling module mime.
    default: Enabling module negotiation.
    default: Enabling module setenvif.
    default: Enabling module filter.
    default: Enabling module deflate.
    default: Enabling module status.
    default: Enabling module reqtimeout.
    default: Enabling conf charset.
    default: Enabling conf localized-error-pages.
    default: Enabling conf other-vhosts-access-log.
    default: Enabling conf security.
    default: Enabling conf serve-cgi-bin.
    default: Enabling site 000-default.
    default: Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/system/apache2.service'.
    default: Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' → '/usr/lib/systemd/system/apache-htcacheclean
service'.
    default: Setting up git (1:2.47.3-0+deb13u1) ...
    default: Processing triggers for man-db (2.13.1-1) ...
    default: Cloning into '/var/www/html'...
ed@fua:~/vagrant-ejemplos$ vagrant ssh
Linux debian 6.12.48+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.48-1 (2025-09-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
vagrant@debian:~$ ls -l /var/www/html/
total 8
-rw-r--r-- 1 root root 2188 Mar 19 18:36 index.html
-rw-r--r-- 1 root root   60 Mar 19 18:36 README.md
vagrant@debian:~$
```



Dirección de los repositorios usados.

Repositorio del vagrant
https://github.com/PabloCarrai/vagrant-ejemplos
Repositorio del html a consumir por la vm
https://github.com/PabloCarrai/html